

Practical Investigations

Edexcel International GCSE Biology (4BI1) — the 14 practical investigations in the specification (numbered by spec point; B = Biology only, not Double Award)

Working scientifically — the essentials

Any valid investigation needs three variable types:

- **Independent variable (IV)** — what you change.
- **Dependent variable (DV)** — what you measure.
- **Controlled variables** — all other factors kept the same, so any change in the DV is caused only by the IV.

Good practice: plan repeats → calculate mean → reduce random error; identify outliers; identify sources of systematic error and how to reduce them.

2.9 — Food tests (glucose, starch, protein, fat)

Method: Grind food with water, filter and split into four portions.

- **Starch:** add drops of iodine → blue-black if starch present.
- **Glucose (reducing sugar):** add Benedict's, heat in a hot (about 80 °C or boiling) water bath for 5 min → green, yellow, orange or brick-red if present (stays blue if absent).
- **Protein:** add biuret reagent (potassium hydroxide, then copper sulfate solution) → blue turns purple if present.
- **Lipid:** dissolve in ethanol, then add water → cloudy white emulsion if present.

Safety: ethanol is flammable — use a water bath, no naked flame.

2.12 — Effect of temperature on enzyme activity (amylase)

Method: Place amylase and starch solutions separately in water baths at 10, 20, 30, 40, 50, 60 °C. Allow to equilibrate. Mix and start timer. Every 30 s, remove a drop onto a spotting tile with iodine. Record time until iodine stays orange-brown (starch fully digested). Rate = $1/\text{time}$. Repeat 3× at each temperature; calculate mean; plot rate vs temperature. Expect an optimum around 37 °C with rate falling sharply above due to denaturation.

2.14B — Effect of pH on enzyme activity

Method: Set up tubes of starch solution, each with a different pH buffer (e.g. pH 3, 5, 7, 9, 11). Put the tubes and a tube of amylase in a water bath at a fixed temperature (e.g. 35 °C) to equilibrate. Add the same volume of amylase to each starch–buffer tube and start the timer. Every 30 s, transfer a drop to a spotting tile containing iodine. Record the time until the iodine no longer turns blue-black. Rate = $1/\text{time}$. Keep temperature, volumes and concentrations of starch and amylase the same. Expect the fastest rate near pH 7 for amylase, with rate falling either side as the active site changes shape (denaturation at extreme pH).

2.17 — Diffusion and osmosis in living and non-living systems

Method: Cut equal potato cylinders with a cork borer. Blot dry, record initial mass. Place one into each of a range of sucrose concentrations (0.0, 0.2, 0.4, 0.6, 0.8, 1.0 mol/dm³) for the same time, at the same temperature. Remove, blot, record final mass. Calculate % change = $((\text{final} - \text{initial}) / \text{initial}) \times 100$. Plot % change vs concentration. The concentration at which % change = 0 is isotonic with the potato cytoplasm.

Typical results: mass increase at low concentrations (water enters by osmosis), mass decrease at high concentrations (water leaves), zero change at the isotonic point (~0.3 mol/dm³).

Method: Non-living system (Visking tubing): tie off a length of Visking (dialysis) tubing, fill it with concentrated sucrose solution, rinse and weigh it, then stand it in distilled water. Re-weigh at intervals: the mass rises as water moves in by osmosis through the partially permeable tubing. For **diffusion**, fill the tubing with a starch and glucose mixture and stand it in water: glucose passes out (Benedict's test on the outside water is positive) but starch molecules are too large (iodine test outside stays negative). A further non-living example: cubes of agar containing an indicator placed in acid — the colour change spreads inwards, and smaller cubes change fully fastest (larger surface area : volume ratio).

2.23 — Photosynthesis: oxygen evolution, starch production and the need for light, CO₂ and chlorophyll

(a) Oxygen evolution from a water plant — effect of light intensity

Method: Place *Elodea* in a test tube of dilute sodium hydrogencarbonate solution (CO₂ supply). Place a lamp at a measured distance. Count oxygen bubbles from the cut stem per minute. Repeat at several distances. Control variables: temperature (use a heat-sink beaker of water), CO₂ concentration, species and length of pondweed. Light intensity $\propto 1/d^2$ (inverse-square law). Plot rate vs $1/d^2$.

(b) Testing a leaf for starch

Method: Destarch the plant in the dark for 24–48 h. Expose to test conditions (e.g. light on half the leaf, dark on the other). Then: boil leaf in water to kill cells; transfer to ethanol in a water bath (*no naked flame*) to remove chlorophyll; rinse in warm water to soften. Add iodine — blue-black where starch is present, yellow-brown where absent.

(c) Showing that light, carbon dioxide and chlorophyll are needed

Method: Destarch the plant first (as above), then test leaves for starch after a few hours in the light.

- **Light:** cover part of a leaf with black paper or foil — only the uncovered part turns blue-black.
- **Carbon dioxide:** enclose one leaf in a sealed bag or flask with soda lime (absorbs CO₂) and a control leaf with no soda lime — the leaf without CO₂ gives no starch.
- **Chlorophyll:** use a variegated leaf (green and white areas); sketch it before testing — only the green areas turn blue-black.

2.33B — Energy content of a food sample

Method: Measure a known volume of water (e.g. 20 cm³) into a boiling tube clamped above the bench and record its starting temperature. Weigh a small piece of dry food (e.g. a crisp or pasta), fix it on a mounted needle, set it alight in a Bunsen flame and hold it immediately under the tube until it stops burning, relighting if necessary. Record the highest temperature. Energy (J) = mass of water (g) × 4.2 × temperature rise (°C), since 1 cm³ of water has a mass of 1 g. Energy per gram = energy ÷ mass of food. Values are much lower than on food labels because heat is lost to the air and glass and the food may not burn completely. Safety: goggles, tie hair back, check for nut allergies.

2.39 — Carbon dioxide and heat from respiring seeds

Method: Heat: soak peas so they germinate; boil a second batch to kill them and rinse both batches in dilute disinfectant (so microbes on dead peas do not respire and warm them). Place each batch in a vacuum flask with a thermometer, plug the neck loosely with cotton wool, and invert the flasks (warm air rises, so heat is not lost through the neck; CO₂ is denser than air and can escape). Record temperatures over several days: the germinating peas warm up, the dead peas do not.

Carbon dioxide: place germinating seeds (or maggots/woodlice on gauze) in a sealed tube above hydrogen-carbonate indicator or limewater, with a control tube of dead seeds or glass beads. Hydrogen-carbonate indicator changes from red/orange to yellow (limewater goes cloudy) as CO₂ is released.

2.45B — Effect of light on net gas exchange in a leaf (hydrogen-carbonate indicator)

Method: Put the same volume of hydrogen-carbonate indicator (red/orange at atmospheric CO₂) into several tubes. Put a leaf into each, held above or in the indicator, and seal with a bung. Place one tube in bright light, one in dim light (e.g. wrapped in gauze) and one in darkness (wrapped in foil); include a tube with no leaf as a control. After about an hour compare colours: **dark** → yellow (respiration only, CO₂ rises); **dim** → stays red/orange (photosynthesis rate = respiration rate, no net exchange); **bright** → purple (photosynthesis faster than respiration, CO₂ falls). The control stays red, showing the leaf causes the change.

2.50 — Breathing in humans: CO₂ release and the effect of exercise

Method: CO₂ in exhaled air: use two boiling tubes of limewater connected so that you breathe in through one tube and out through the other. The limewater that exhaled air bubbles through turns cloudy much faster, showing exhaled air contains more CO₂ than inhaled air.

Effect of exercise: sitting still, count breaths per minute (breathing rate) for several minutes to get a resting mean. Do a set exercise (e.g. step-ups for 2 minutes), then immediately count breaths per minute, and every minute until the rate returns to resting. Breathing rate (and depth) rises during exercise to supply more oxygen and remove extra CO₂ from faster respiration. Control the type, intensity and duration of exercise; use several people to allow for individual variation.

2.58B — Environmental factors and the rate of transpiration (potometer)

Method: Cut a leafy shoot under water to avoid air bubbles. Assemble potometer under water and attach shoot. Dry leaves. Record the rate of movement of an air bubble in the capillary tube over time — this approximates the rate of water uptake $\approx$ transpiration rate. Change one factor at a time (temperature, humidity, wind speed, light intensity) — e.g. a fan for wind, a plastic bag for humidity, a lamp at different distances for light. Record and repeat. Rate increases with higher temperature, wind speed and light intensity, and decreases with higher humidity.

3.5 — Conditions needed for seed germination

Method: Set up tubes each containing the same number of seeds (e.g. cress or mung beans), differing in one condition at a time: **complete** — damp cotton wool, air, warm (about 20 °C); **no water** — dry cotton wool; **no oxygen** — seeds in boiled, cooled water with a layer of oil on top (or a sealed flask with pyrogallol, handled with care); **cold** — damp cotton wool in a fridge (about 4 °C). Leave for several days and record the number (percentage) of seeds germinated in each. Germination only occurs reliably with water, oxygen and a suitable temperature — light is not needed (test with a tube kept in the dark).

4.2 — Population size in two different areas using quadrats

Method: Lay out an area. Generate random coordinates. Place a 0.5 m × 0.5 m quadrat at each. Count individuals of chosen species (or estimate % cover). Repeat 10+ times. Calculate mean per quadrat, multiply by total area ÷ quadrat area to estimate population. Repeat in the second area (e.g. a mown and an unmown field, or shaded and open ground) using the same quadrat size, number of quadrats and sampling method, then compare the two estimates.

4.4B — Distribution of organisms and measuring biodiversity using quadrats

Method: To study **distribution**, run a tape (transect) across a habitat where conditions change, e.g. from a hedge into open field or down a rocky shore. Place a quadrat at regular intervals along the line and record which species are present, their number or percentage cover, and a physical factor (e.g. light intensity, soil moisture) at each point. To measure **biodiversity**, record the number of different species (species richness) and how many individuals of each (evenness) in randomly placed quadrats in each habitat; more species with more even numbers means higher biodiversity. Keep quadrat size and number, and identification method, the same.

5.6 — Anaerobic respiration by yeast in different conditions

Method: Mix yeast with sugar solution in a test tube; cover the surface with a thin layer of oil to keep oxygen out (after first boiling and cooling the water to remove dissolved oxygen). Connect a delivery tube into limewater or hydrogen-carbonate indicator, or collect gas in an upturned measuring cylinder / gas syringe. Stand the tube in a water bath and count bubbles or measure the volume of CO₂ produced per minute. Change one condition at a time: temperature (e.g. 20–60 °C), type or concentration of sugar. Rate of CO₂ production rises with temperature up to an optimum (around 35–40 °C), then falls as enzymes denature. Anaerobic respiration in yeast: glucose → ethanol + carbon dioxide.

Not prescribed practicals for 4BI1 — useful supporting methods

The methods below are not practical investigations in the 4BI1 specification, but they support the topics above.

Measuring oxygen uptake with a respirometer (supports 2.39)

Method: Use a respirometer: germinating peas in a boiling tube with soda lime (absorbs CO_2) below a gauze. Seal with a bung and capillary tube dipped into coloured water. Temperature controlled by water bath. As peas respire aerobically, O_2 is consumed (CO_2 is absorbed by soda lime), gas volume falls and coloured water moves up the capillary. Rate = distance moved / time. Control with boiled (dead) peas. Repeat for reliability.

Making a microscope slide of plant cells (supports 2.2 cell structure)

Method: Peel a thin epidermal layer from an onion. Place flat on a slide in a drop of water. Add a drop of iodine (stain). Lower a coverslip at 45° with a mounted needle to avoid bubbles. Blot excess. View under $\times 40$, then $\times 100$, then $\times 400$ total magnification. Record actual size using eyepiece graticule and stage micrometer; calculate magnification.

Exam Tip: When Edexcel asks you to "describe an experiment", always include: IV, DV, control variables, method, repeats (reliability), and a comment on how to reduce error. Missing one of these costs easy marks.